

Task 2: Data Visualization and Storytelling Report

Project Title: Superstore Sales Performance Dashboard and 15-Day Sales Forecast using Power BI

Submitted by: Himani Khawashi

Tool Used: Microsoft Power BI

Dataset: Superstore Sales Dataset

1. Objective

The objective of this project is to analyze Superstore sales data and create an interactive dashboard that effectively communicates business performance through data visualization and storytelling. The dashboard enables decision-makers to monitor key performance indicators (KPIs), identify sales trends, compare regional performance, and forecast future sales for better planning.

2. Dataset Description

The project uses the **Superstore Sales Dataset**, which contains historical sales transactions including:

- Order Date
- Ship Date
- Category
- Sub-Category
- Segment
- Region
- State
- Sales
- Profit
- Quantity
- Payment Mode
- Ship Mode

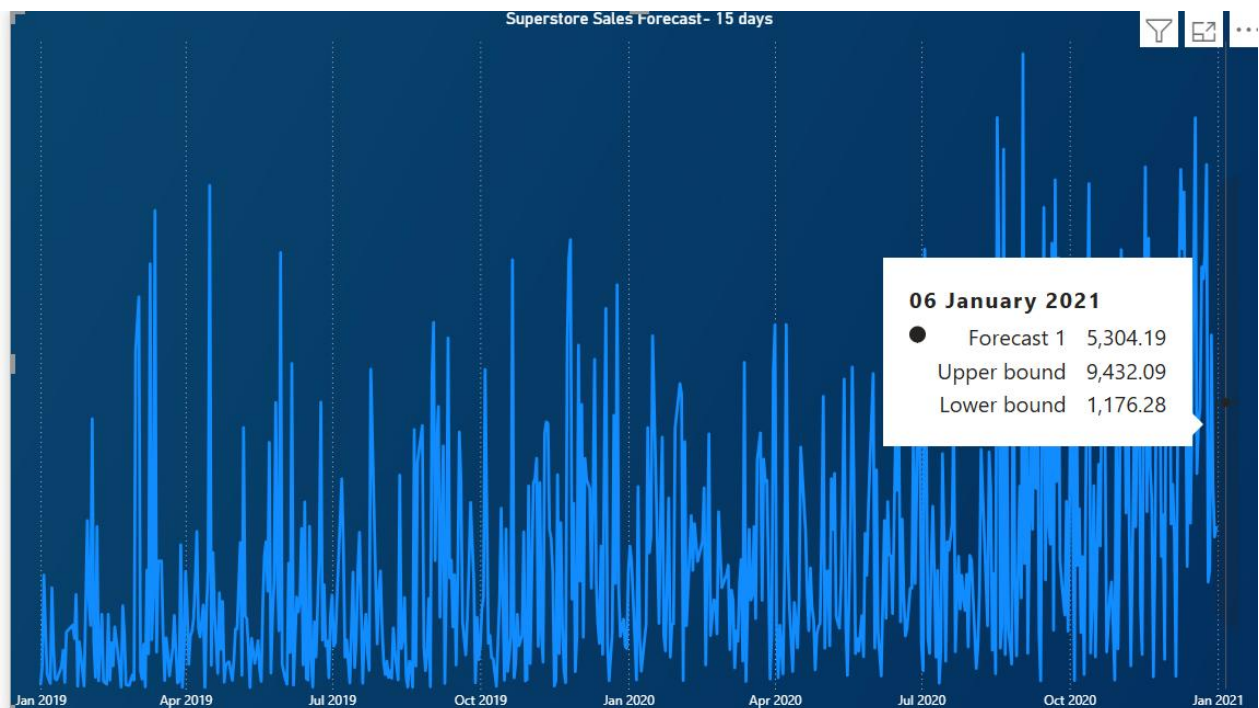
The data was analyzed and visualized using **Microsoft Power BI**.

3. Dashboard:

Page 1: Sales Performance Dashboard



Page 2: Sales Forecast Dashboard



4. Dashboard Overview

The dashboard consists of **two interactive pages**.

Page 1 – Sales Performance Dashboard

This page provides a complete overview of the company's sales performance through interactive KPIs and visualizations.

Key Performance Indicators (KPIs)

- **Total Sales:** 1.57 Million
- **Total Quantity Sold:** 22 Thousand Units
- **Average Delivery Time:** 4 Days
- **Total Profit:** 175.2 Thousand

Visualizations

- Monthly Sales Trend
- Monthly Profit Trend
- Sales by Category
- Sales by Sub-Category
- Sales by Region
- Sales by State (Map)
- Sales by Segment
- Sales by Ship Mode
- Sales by Payment Mode
- Region Slicer for interactive filtering

Page 2 – Sales Forecast Dashboard

The second page presents a **15-Day Sales Forecast** based on historical sales data.

The forecast helps estimate future sales patterns, enabling management to plan inventory, logistics, and marketing strategies proactively.

5. Business Storytelling

The dashboard reveals a clear business story about the Superstore's sales performance.

The company generated **1.57 million** in total sales while earning a profit of approximately **175 thousand**. Around **22,000 products** were sold, with an average delivery time of **4 days**, indicating efficient order fulfillment.

Sales show a steady upward trend throughout the year, with significant growth during the last quarter. This suggests increased customer demand during the festive and holiday season.

Among product categories, **Office Supplies** generated the highest sales, followed closely by **Technology** and **Furniture**. Within sub-categories, **Phones** emerged as the best-selling product.

The **West region** contributed the highest percentage of total sales, followed by the East region, indicating strong market performance in these areas.

Customers preferred **Standard Class** shipping and mostly selected **Cash on Delivery (COD)** as their payment method.

The forecasting model predicts that sales are expected to remain stable with occasional spikes over the next **15 days**, helping management prepare inventory and operational resources in advance.

6. Key Insights

- Total Sales reached **1.57 Million**.
- Total Profit was approximately **175 Thousand**.
- Average Delivery Time is **4 Days**, reflecting efficient logistics.
- Office Supplies generated the highest revenue.
- Phones are the top-performing sub-category.
- West region recorded the highest sales contribution.
- Standard Class is the preferred shipping method.
- COD is the most frequently used payment mode.
- Sales increased noticeably during the final months of the year.
- Forecast indicates continued demand over the next 15 days.

7. Business Recommendations

- Increase inventory for high-performing categories such as Office Supplies and Phones.
- Focus marketing campaigns on the West and East regions to maximize revenue.

- Improve promotional strategies in lower-performing regions.
- Continue optimizing delivery operations to maintain the current 4-day average delivery time.
- Utilize sales forecasting for inventory management and workforce planning during high-demand periods.

8. Conclusion

The Power BI dashboard successfully transforms raw sales data into meaningful business insights using interactive visualizations. It provides an easy-to-understand overview of sales performance, customer preferences, regional trends, and operational efficiency.

The addition of a **15-Day Sales Forecast** enhances the dashboard by supporting future planning and enabling proactive business decisions. Overall, this project demonstrates how data visualization and storytelling can help organizations make informed, data-driven decisions.